Muhammed Abdul Quadir Owais

Ex-Software Engineer | AWS AI cert. | AWS |DSA, ML Research ENTHUSIAST | CONSISTENT LEARNER
California, USA | quadirowais2000@gmail.com | +1 (805)-803-2614 | **LeetCode** (570+) | **LinkedIn** | **GitHub** | **Portfolio** | **Kaggle**

SUMMARY:

Software Engineer with **5+ years** of programming and **1.5+ years** of applied ML research experience with **3+ in-progress publications**. Strong background in **software system design, full-stack development** and **building applications, developing** and optimizing **AWS pipelines (MLOps), test pipelines,** and resolving production issues**.** Proven ability to deliver scalable, maintainable, and robust solutions within large cross-domain teams, contributing to multi-million-dollar revenue.

EDUCATION

California State University, Channel Islands **CA, USA | Jan '24 - Dec'25**
Master of Science (Computer Science) ***CGPA - 3.9/4.0***
Courses: Master's Thesis (StatGenAI), AI, Neural Networks, Matrix Manifolds in ML, Adv. Algo., Cloud Computing AWS

University College of Engineering, Osmania University **HYD, IND | Jun '18 - May '22**
Bachelor of Engineering (B.E.) in Electronics and Communication Engineering (ECE)

EXPERIENCE

Software Engineer II - THOUGHTSPOT (Business Intelligence AI company**)** **HYD, IND | Aug '22 - Sept '23**

Tech.: Java | Python | Javascript |Typescript | ReactJs | SpringBoot | GraphQL | CI/CD pipelines | LSTMs

- Developed a new Homepage and Navigation system with GraphQL, improving response time by 15% and increasing feature adoption by 10% over **microservices architecture.** Added testing pipelines for Teams edition to improve the product quality. A/B testing on various components to gauge user engagement. Mentored junior engineers by establishing best practices for scalable architecture, efficient code, and test-driven development

Research Assistant - GenAI Virtual Reality - CSUCI (Prof. B. Abbasi & K. Linton**) (Endorsement) Jul '24 - Jan'25**

Tech.: C# | Unity | Agentic AI| LLMs | LangChain | HuggingFace | Meta | OpenAI APIs | Speech2Text – Text2Speech

- Helping brain injury patients with ML & Virtual scenarios to regain abilities like memory, focus, and communication.
- Phase 1: Resolved cross-platform, made HTC's Focus and Memory Scenarios meta-compatible, and **saved 60% time** in setup for brain injury patients. **80% of patients** got better with VR therapy. (Manuscript Link)
- Phase 2: Communication Scenario using GenAI models and experimenting with custom-made models.
- Designed and developed AI agent workflows in Python, implementing modular multi-agent architectures. Built specialized agents including Analyst Agents, Configuration/Creation Agents, Manager Agents, and Skill-based Data Fetching Agents. Developed Orchestrator and Router agents to manage task delegation, inter-agent communication, and workflow optimization. Optimized agent performance through structured prompt design, tool integration, and workflow monitoring.

Research Thesis Assistant - CSUCI (Advisors: Prof. Micheal Solty & Dr. Ricardo Suarez**)** **Jan '24 - Dec'25**

Tech.: Django| AWS (EC2, CodeDeploy, S3, CI-CD, SQS, Lambda)| ML models | Math. dist. | VAEs | KDE | Matrix Manifold

- Researching the machine learning architectures with the synergy of **GANs, CTGANS, VAEs, KDE,** and **MVGD distributions** and verifying with clustering algorithms like K-means and Gaussian Mixture Model. Thus, synthetic data is generated up to **15 dimensions (features) up to 99% accuracy.** Currently optimizing to enhance accuracy, scalability, and dimensionality with advanced mathematical methods including **Matrix Manifolds. (**Draft_ work**, Thesis)**
- Deployed web app over AWS and saved the synthetically generated data on the S3 using lambda, commit pipeline to the EC2 from GitHub using AWS CodeDeploy.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, C#, JavaScript
- **Software Engineering:** ReactJs, Django, SpringBoot, GraphQL, AWS, Git, CI/CD pipelines, Unity, Django, MongoDB, Redis, Gradle
- **MACHINE LEARNING:** Pytorch | TensorFlow | Keras | NumPy | Pandas | Scikit-learn | LangChain | Classical ML models | OpenAI API | LLMs | SLM | Hugging Face | CNNs | RNNs | LSTMs | GANs | VAEs | Hyperparameter tuning | cross-validation | Feature Engineering | Literature Review

PUBLICATIONS:

Virtual Reality Rehabilitation for a Post-ICU Hospitalized Patient with a Traumatic Brain Injury: A Case Report
Neuropsychological Rehabilitation – (Under review) (Manuscript Link)

AIENs: Adaptive Initializer Enhancer Networks : (For synthetic data generation) (IN PROGRESS)

- Uses Adaptive kernel density estimation (AKDE) in place of the Gaussian distribution while calculating the KL divergence loss, with modifications in the VAE architecture, 19% reduced learning time, hence less computations, as the data pattern is already captured in the AKDE prior, and 9% data fidelity.

Manifold-based Data Generation: (IN PROGRESS)

- This study introduces generative models that operate on matrix manifolds, aiming to capture the intrinsic patterns of data. The approach seeks to enhance the fidelity of synthetic data by respecting the underlying manifold topology. (And exploring Matrix Manifolds to leverage the Machine learning status quo)

Projects (Software Engineering)

QEATS - ***Food Ordering App*** **(CODE LINK)** **Jan - Feb '22**

Skills: Java| Spring Boot|REST API| Jackson| Mockito| MongoDB| Redis | JMeter| MongoDB query | Multithreading.

- Implemented **MVCS** framework with 7+ **REST API** endpoints to query restaurant information and place orders.
- Boosted the app performance by 19%, using **Redis Caching** to retrieve the repeated data from **MongoDB** and **multithreading** to run *parallel search queries* for the search support feature (to map a list of restaurants).
- Addressing the real-world problems and production issues using **Scientific Debugging** methods and the **large load scenarios** using the JMeter tool, and hitting 100 to 1000 API requests/sec.

Open source Development – ***Sympy*** **(CODE LINK)** **Feb - Apr '19 & Nov '20 - May '21**

Skills: Python| Version Control System → GitHub - Commits, merges | Code review | Math | Lambert Equations.

- Did **16 merges,** 6 open, and 10 closed pull requests, along with writing test cases.
- Debugged the issues**, code review**, and did work left by others on the SYMPY platform, mostly on the Lambert equation.

Web Development – ***News website*** **(CODE LINK)** **Mar-May '20**

Skills: Python| Django | REST APIs | HTML | CSS | JQUERY | JAVASCRIPT | BOOTSTRAP | CDN | Googling | Tutorial.

- Developed the news website with **5+ features** like *Authentication* (Log-in/out), *Share opt*ion, *Searching blogs/items*, Posting Comments, Blogs publishing (WYSIWYG summer note) by admin, adding geoIp2 (get geolocation of user).
- Connected with PostgreSQL DB, and a **responsive design** using front-end technologies.

QMoney - ***Visual Stock Portfolio Analyzer*** **(CODE LINK)** **Nov - Dec '21**

Skills: Java| REST APIs| Jackson| Interfaces | Code Refactoring| Gradle| Multithreading.

- Fetched the data from 2 **real-time data APIs** (i.e., tiingo, alphavantage) for stocks and analyzed the business logic to calculate the annual returns with the restTemplate and Jackson library, and gave the visual insights of stock return.
- Improved 17% boost in performance using the concept of **multithreading**.

Object Modeling – ***JukeBox and Contest platform Modeling*** **Dec '21 - Jan '22**

Skills: Java| Object Oriented Design| Design Patterns| SOLID Principles | Low-level Design | Unit Testing| TDD.

- Modeled a jukebox, contest platform by applying your **Core Java + design skills** from the problem statement.
- Designed and implemented the business logic & the classes required, beginning with low-level Design and **UML diagrams.**